INTERNATIONAL BACCALAUREATE DIPLOMA PROGRAMME

IB DP Biology

Complete Cheat Sheet

Full Syllabus · HL + SL · Score a 7

20 pages of premium revision content

TOPICS COVERED

- 🔬 Cell Biology
- 🧬 Molecular Biology
- 🧪 Genetics
- 🌿 Ecology
- 🌱 Evolution
- 🫀 Human Physiology
- ⭐ HL Extensions
- 📝 Exam Mastery



WHAT'S INSIDE

- ✦ All 6 core topics
- ✦ HL extension content
- ✦ Labeled diagrams
- ✦ Flowcharts & cycles
- ✦ Exam tips & warnings
- ✦ Command terms guide
- ✦ Key definitions
- ✦ Quick recall boxes
- ✦ Common mistakes
- ✦ Night-before checklist

SCORING TARGET

7/7

IB Biology Grade

Based on current IB Biology syllabus (2025 onwards)

GETTING STARTED

How to Use This Cheat Sheet

First Read

Skim all sections. Note topics you're weak on. Use the syllabus map to track coverage.

Active Recall

Cover the page. Recite definitions out loud. Return to "Quick Recall" boxes often.

Exam Mode

Use Pages 16–20 the night before. Focus on Exam Tips and Command Terms.

⚡ **EXAM TIP:** Study diagrams actively — draw them from memory. IB often asks for labeled diagrams worth 3–5 marks.

SYLLABUS ROADMAP

Topic 1 — Cell Biology PAGES 3–4

- Cell theory & cell types
- Organelles & functions
- Membrane structure & transport
- SA:V ratio

Topic 2 — Molecular Biology PAGES 5–6

- Biomolecules: carbs, lipids, proteins
- Enzymes & factors
- DNA structure & replication
- Transcription & translation

Topic 3 — Genetics PAGES 7–8

- Genes, alleles, dominance
- Punnett squares & crosses
- Pedigree analysis
- Sex-linked traits

Topic 4 — Ecology PAGE 9

- Food chains & webs
- Energy flow & pyramids
- Carbon & nitrogen cycles
- Population dynamics

Topic 5 — Evolution & Biodiversity PAGE 10

- Natural selection mechanism
- Speciation (allopatric/sympatric)
- Classification system
- Cladistics

Topic 6 — Human Physiology PAGES 11–13

- Digestion & absorption
- Cellular respiration (aerobic/anaerobic)
- Circulation & blood
- Immunity & defence

HL Extensions HL ONLY Pages 14–15

- Nucleic acids (deeper) · Metabolism · Plant biology

Exam Mastery Pages 16–20

- Mistakes · Strategy · Definitions · Quick Revision · Checklist

💡 **QUICK RECALL:** Look for ⚡ Exam Tips, ⚠ Mistakes, 💡 Quick Recall boxes, and HL labels throughout.

📌 Cell Theory (3 Tenets)

- All living organisms are composed of **cells**
- Cells are the **basic structural and functional unit** of life
- All cells arise from **pre-existing cells** (cell division)

> ⚡ **EXAM TIP:** IB often asks you to "state" cell theory — use exact wording above.

⚡ Prokaryote vs. Eukaryote

Feature	Prokaryote	Eukaryote
Nucleus	❌ (nucleoid)	✅
Size	1–10 µm	10–100 µm
Membrane-bound organelles	❌	✅
Ribosomes	70S	80S (cytoplasm)
DNA	Circular, naked	Linear + histone
Cell wall	Peptidoglycan	Varies/absent

🏗️ Organelles & Functions

Organelle	Function	Organelle	Function
Nucleus	Contains DNA; controls cell activity	**Mitochondria**	Aerobic respiration; ATP production
Ribosome	Protein synthesis (translation)	**Chloroplast**	Photosynthesis (plants/algae only)
RER	Synthesis of secretory/membrane proteins	**Lysosome**	Digestion via hydrolytic enzymes
SER	Lipid synthesis; detoxification	**Vacuole**	Storage; turgor pressure (plants)
Golgi apparatus	Modifies, packages, secretes proteins	**Centriole**	Cell division (animal cells)

📐 Surface Area : Volume Ratio

- **SA:V decreases** as cell size increases
- Small cells = high SA:V = efficient exchange
- Large cells = low SA:V = inefficient → cell divides

> ⚡ **EXAM TIP:** If asked "why must cells remain small" → SA:V ratio argument + efficient diffusion distances.

Cube formula:
SA = 6 × side²
V = side³
SA:V = 6/side

✨ Emergent Properties + Exceptions

EMERGENT PROPERTIES

- Multicellular organisms have properties not found in individual cells
- E.g. consciousness, immune response

EXCEPTIONS TO CELL THEORY

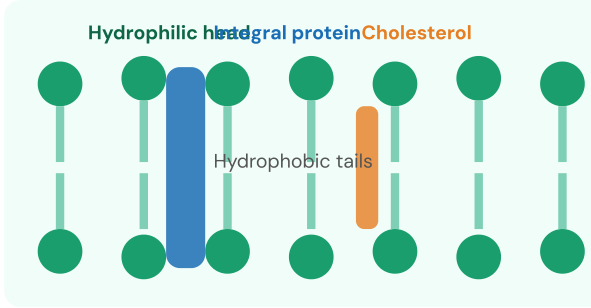
- **Striated muscle** — multinucleated (syncytium)
- **Aseptate fungi** — no cross walls
- **Giant algae** (e.g. Acetabularia) — single huge cell
- **Viruses** — not made of cells

> ⚠ **COMMON MISTAKE:** Ribosomes are NOT membrane-bound organelles — they are free in cytoplasm or on RER surface.

🔬 SECTION 1 — CELL BIOLOGY: MEMBRANE & TRANSPORT

🌊 Fluid Mosaic Model of the Membrane

- **Phospholipid bilayer** — hydrophilic heads face outward; hydrophobic tails face inward
- **Fluid** — phospholipids can move laterally; membrane is not rigid
- **Mosaic** — proteins embedded throughout
- **Integral proteins** — span full bilayer (e.g. channels, pumps)
- **Peripheral proteins** — on surface only
- **Cholesterol** — stabilises fluidity across temperatures
- **Glycoproteins/Glycolipids** — cell recognition, signalling



Fluid Mosaic Model (simplified)

⚡ Membrane Transport

Diffusion (Passive)

- High → low conc
- No ATP needed
- Small, nonpolar molecules (O_2, CO_2)
- ↑ temp, ↑ SA, ↑ gradient → faster

Osmosis (Passive)

- Water movement via semipermeable membrane
- Low water potential → high water potential
- **Isotonic** — no net movement
- **Hypertonic** — cell shrinks (crenation)
- **Hypotonic** — cell swells/lysates

Active Transport

- Low → high conc (against gradient)
- Requires ATP + carrier proteins
- Example: Na^+/K^+ pump
- 3 Na^+ out, 2 K^+ in per ATP

Endocytosis & Exocytosis

- **Endocytosis** — membrane engulfs particles (in)
- **Phagocytosis** — solid particles (e.g. bacteria)
- **Pinocytosis** — liquid droplets
- **Exocytosis** — vesicles fuse and release (out)
- Both require ATP; bulk transport

⚡ **EXAM TIP:** Osmosis is diffusion of WATER only — not solutes. Water moves to where there are MORE solutes (lower water potential).

⚠ **COMMON MISTAKE:** Active transport doesn't just mean "uses energy" — it specifically moves substances AGAINST their concentration gradient.

💡 **QUICK RECALL:** Passive transport: no ATP (diffusion, osmosis, facilitated diffusion). Active: ATP required (active transport, endo/exocytosis).

🍞 Carbohydrates

MONOMERS → POLYMERS

- **Monosaccharides** (e.g. glucose, fructose, ribose)
- **Disaccharides** (e.g. maltose, sucrose, lactose)
- **Polysaccharides** (starch, glycogen, cellulose, chitin)

FUNCTIONS

- Energy source (glucose → ATP)
- Energy storage (starch/glycogen)
- Structural (cellulose in plants)

Condensation = bond formed + H_2O released
Hydrolysis = bond broken + H_2O added

🫙 Lipids

TYPES

- **Triglycerides** — 1 glycerol + 3 fatty acids; energy storage
- **Phospholipids** — glycerol + 2 FA + phosphate; membrane
- **Steroids** — cholesterol, sex hormones

FUNCTIONS

- Long-term energy storage
- Insulation & protection
- Membrane structure (phospholipids)
- Hormone production

Saturated FA = no double bonds (solid at room temp)
Unsaturated FA = double bonds (liquid/oil)

🔧 Proteins

STRUCTURE LEVELS

- **Primary** — amino acid sequence (peptide bonds)
- **Secondary** — α-helix / β-pleated sheet (H-bonds)
- **Tertiary** — 3D folding (ionic, H, disulfide bonds)
- **Quaternary** — 2+ polypeptides (e.g. haemoglobin)

FUNCTIONS

- Enzymes, antibodies, transport (Hb)
- Structural (collagen, keratin)
- Hormones (insulin)
- Receptors & channel proteins

⚙️ Enzymes

KEY CONCEPTS

- **Biological catalysts** — lower activation energy
- **Active site** — specific shape for substrate
- **Enzyme-substrate complex** forms → products released
- **Lock and Key** vs **Induced Fit** (active site changes shape slightly)
- Enzymes are **not consumed** in reaction
- Denaturation = irreversible loss of shape (high temp or extreme pH)

FACTORS AFFECTING ENZYME ACTIVITY

Factor	Effect on Rate	Optimum
Temperature	↑ until denatured	~37°C (human)
pH	Specific optimum	pH 7 (most)
[Substrate]	↑ until Vmax	All sites full
[Enzyme]	↑ proportionally	No max (if substrate excess)

INHIBITION

- **Competitive** — blocks active site; ↑ substrate overcomes it
- **Non-competitive** — binds allosteric site; changes active site shape; irreversible

⚡ **EXAM TIP:** When asked to explain enzyme action using "induced fit" — describe the active site as changing shape to fit the substrate, increasing binding specificity.

⚠ **COMMON MISTAKE:** Enzymes lower activation energy — they do NOT change the energy of reactants or products (ΔG is unchanged).

🧬 SECTION 2 — MOLECULAR BIOLOGY: DNA & GENE EXPRESSION

🔗 DNA Structure

- **Double helix** — 2 antiparallel strands
- **Nucleotide** = deoxyribose + phosphate + nitrogenous base
- **Base pairs**: A=T (2 H-bonds), G≡C (3 H-bonds)
- **Backbone**: sugar-phosphate (covalent bonds)
- **5'→3' directionality** on each strand
- Complementary strands are antiparallel

Purines (double ring): Adenine, Guanine
Pyrimidines (single ring): Thymine, Cytosine (Uracil in RNA)

🔄 DNA Replication (Semi-Conservative)

1. **Helicase** unwinds and separates DNA strands (breaks H-bonds)
2. **Primase** adds RNA primer to provide 3'–OH start
3. **DNA polymerase III** adds nucleotides 5'→3' (reads template 3'→5')
4. **Leading strand** — continuous synthesis; **Lagging strand** — Okazaki fragments
5. **DNA polymerase I** removes primers; **Ligase** joins fragments

⚡ **EXAM TIP:** Result: 2 identical double-stranded DNA molecules, each with 1 original + 1 new strand.

📡 Transcription (DNA → mRNA)

1. **RNA polymerase** binds to promoter region
2. DNA unwinds; template strand read 3'→5'
3. Complementary RNA nucleotides added (U replaces T)
4. mRNA released; DNA rewinds

MRNA PROCESSING (EUKARYOTES)

- **Introns** removed (non-coding)
- **Exons** joined (coding)
- **5' cap + poly-A tail** added for stability
- mRNA exits via nuclear pore to ribosome

HL Alternative splicing allows one gene to produce multiple proteins

🏭 Translation (mRNA → Protein)

1. **Initiation**: ribosome binds mRNA at start codon (AUG → Met)
2. **Elongation**: tRNA anticodon matches mRNA codon; peptide bond forms
3. Ribosome moves along mRNA (5'→3'); polypeptide grows
4. **Termination**: stop codon (UAA, UAG, UGA) reached; chain released

KEY PLAYERS

- **mRNA** — carries code (codons)
- **tRNA** — carries amino acid; has anticodon
- **rRNA** — ribosome structure + peptidyl transferase activity
- **Ribosome sites**: A (aminoacyl), P (peptidyl), E (exit)

Codon = 3 bases on mRNA
Anticodon = 3 bases on tRNA
64 codons — 61 code AA + 3 stop

Central Dogma: DNA → mRNA → Protein (Transcription → Translation)

⚠ **COMMON MISTAKE:** tRNA has the ANTICODON, not the codon. The codon is on mRNA. Don't confuse them in exam answers.

Essential Genetics Vocabulary

Gene	A sequence of DNA that codes for a functional product (protein or RNA)	**Dominant**	Allele expressed in heterozygous state (capital letter)
Allele	A variant of a gene; different forms at the same locus	**Recessive**	Expressed only when homozygous (lowercase)
Genotype	The genetic makeup (allele combination, e.g. Aa)	**Codominance**	Both alleles fully expressed (e.g. ABO blood type)
Phenotype	The observable characteristics (expressed genotype)	**Locus**	Specific position of a gene on a chromosome
Homozygous	Two identical alleles (AA or aa)	**Chromosome**	Linear DNA molecule with associated proteins (histones)
Heterozygous	Two different alleles (Aa)	**Diploid/Haploid**	2n (pairs) / n (single set) — humans: 2n=46

Types of Dominance

Type	Example	Phenotype
Complete dominance	Aa (brown eye)	Only dominant shown
Codominance	ABO blood type	Both alleles shown
Incomplete dominance	Red × White flower	Intermediate (pink)
Sex-linked	Colour blindness	X-linked traits

ABO Blood Types:

I^AI^A or I^Ai = A | I^BI^B or I^Bi = B

I^AI^B = AB (universal recipient) | ii = O (universal donor)

Meiosis Summary

- Produces 4 **haploid** genetically unique cells
- **Meiosis I** — homologous chromosomes separate
- **Meiosis II** — sister chromatids separate
- **Crossing over** (chiasmata) in Prophase I — genetic recombination
- **Independent assortment** — random orientation at metaphase I
- Sources of genetic variation: crossing over + independent assortment + random fertilisation

> ⚡ **EXAM TIP:** IB asks: "Explain how meiosis produces genetic variation" → State 3 mechanisms above with detail.

Punnett Squares — Worked Examples

MONOHYBRID CROSS (AA × AA)

	A	a
A	AA	Aa
a	Aa	aa

Ratio: **3 dom : 1 rec**

TEST CROSS (AA × AA)

	A	a
a	Aa	aa
a	Aa	aa

Ratio: **1 dom : 1 rec**

SEX-LINKED (CARRIER FEMALE × NORMAL MALE)

	X^N	Y
X^N	X^NX^N	X^NY
X^n	X^NX^n	X^nY

50% sons affected; 50% daughters carriers

> ⚠ **COMMON MISTAKE:** When doing a sex-linked cross, always write the allele on the X chromosome (X^N), not just letters alone — IB requires proper notation.

SECTION 3 — GENETICS: APPLICATIONS & PEDIGREES

Pedigree Analysis

SYMBOLS

Female ○ Male □ Affected ♀ ○ Affected ♂ □

DETERMINING INHERITANCE PATTERN

- **Autosomal dominant**: trait appears every generation; affected parent always present
- **Autosomal recessive**: can skip generations; both parents unaffected can have affected child
- **X-linked recessive**: more males affected; never father→son transmission; carrier mothers
- **X-linked dominant**: affected father passes to ALL daughters

Mutations

GENE MUTATIONS

- **Substitution** — 1 base replaced; may be silent, missense, or nonsense
- **Insertion/Deletion** — frameshift; affects all downstream codons
- Sickle cell anaemia = point substitution (GAG→GTG); valine replaces glutamic acid

CHROMOSOME MUTATIONS

- **Non-disjunction** — chromosomes fail to separate in meiosis
- **Trisomy 21** (Down syndrome) — extra chr 21
- Mutagens: UV radiation, X-rays, chemical mutagens

> **EXAM TIP:** Frameshift mutations are more severe than substitutions — explain why (all codons downstream are affected).

Polygenic Traits

- Controlled by **two or more genes**
- Shows a **continuous distribution** (bell curve)
- Examples: skin colour, height, intelligence
- Influenced by both genes AND environment
- Cannot draw simple Punnett squares

Genetic Biotechnology (SL Overview)

- **PCR** — amplifies specific DNA sequences; needs primers, Taq polymerase, thermocycler
- **Gel electrophoresis** — separates DNA by size; smaller fragments move further
- **Genetic modification** — inserting genes into host organisms (GMOs)
- **CRISPR** — targeted gene editing using guide RNA

HL Restriction enzymes create sticky ends; DNA ligase seals recombinant DNA

Common IB Genetics Question Types

CALCULATION

- Genotype/phenotype ratios from crosses
- Probability of offspring genotype
- Chi-squared test for genetics data

DIAGRAM

- Complete a Punnett square
- Interpret a pedigree chart
- Identify gametes correctly

EXPLAIN/DESCRIBE

- Determine mode of inheritance
- Explain genetic variation in meiosis
- Analyse mutation effects

> **QUICK RECALL:** Pedigree tip: If two unaffected parents have an affected child → trait must be autosomal RECESSIVE.

Ecological Organisation

Organism → Individual living thing

Population → All individuals of same species in area

Community → All populations in same area

Ecosystem → Community + abiotic environment

Biosphere → All ecosystems on Earth

Autotrophs (producers) — make their own food via photosynthesis
Heterotrophs (consumers/decomposers) — obtain energy from others

Food Chains & Webs

TROPHIC LEVELS

- **TL1 — Producers**: Plants, algae (photosynthesis)
- **TL2 — Primary consumers**: Herbivores
- **TL3 — Secondary consumers**: Carnivores
- **TL4 — Tertiary consumers**: Top predators
- **Decomposers**: Bacteria, fungi (break down dead organic matter)

ENERGY TRANSFER

- Only ~**10% of energy** transferred between trophic levels
- 90% lost as heat, excretion, respiration
- Energy flows ONE way; cannot be recycled
- Matter (nutrients) CAN be recycled via cycles

> ⚡ **EXAM TIP:** Energy pyramids are always pyramid-shaped. Biomass pyramids may invert (aquatic). Number pyramids may invert (parasites).

Carbon Cycle

KEY PROCESSES

- **Photosynthesis** — removes CO_2 from atmosphere (fixes carbon)
- **Respiration** — releases CO_2 (all organisms)
- **Combustion** — fossil fuels release CO_2
- **Decomposition** — microbes release CO_2 from dead matter
- **Calcification** — CO_2 stored in shells/limestone
- **Ocean absorption** — dissolves CO_2 (carbonic acid)

> ⚠ **COMMON MISTAKE:** Human impact: burning fossil fuels increases atmospheric CO_2 → greenhouse effect → climate change.

Nitrogen Cycle

KEY PROCESSES & BACTERIA

- **Nitrogen fixation**: N_2 → NH_3 (Rhizobium in root nodules; Azotobacter)
- **Nitrification**: NH_3 → NO_2^- → NO_3^- (Nitrosomonas → Nitrobacter)
- **Assimilation**: Plants absorb NO_3^-; used for amino acids/nucleotides
- **Decomposition (ammonification)**: Organic N → NH_3/NH_4^+
- **Denitrification**: NO_3^- → N_2 (Pseudomonas; anaerobic conditions)

> ⚡ **EXAM TIP:** Name specific bacteria in nitrogen cycle — examiners reward precision (Rhizobium, Nitrosomonas, Nitrobacter, Pseudomonas).

Population Dynamics

J-CURVE (EXPONENTIAL)

- Rapid growth; no limiting factors
- Unlimited resources assumed
- Example: bacteria in new medium

S-CURVE (LOGISTIC)

- Growth slows as K (carrying capacity) approached
- Limiting factors: food, space, predation
- Most natural populations follow S-curve

FACTORS AFFECTING SIZE

- **Density-dependent**: predation, disease, competition
- **Density-independent**: climate, natural disasters

Natural Selection — Mechanism

1. **Variation** exists within a population (heritable)
2. **Overproduction**: more offspring than environment can support
3. **Struggle for existence**: competition for limited resources
4. **Differential survival**: individuals with advantageous traits more likely to survive
5. **Reproduction**: survivors pass advantageous alleles to offspring
6. **Allele frequency** of advantageous traits increases over generations

> ⚡ **EXAM TIP:** IB asks to "explain the mechanism of natural selection" — use all 6 steps above. Don't write "survival of fittest" as it's imprecise.

Evidence for Evolution

- **Fossil record** — shows gradual change over time; transitional forms
- **Homologous structures** — same origin, different function (e.g. vertebrate limbs)
- **Analogous structures** — same function, different origin (convergent evolution)
- **Vestigial structures** — non-functional remnants (e.g. appendix, whale pelvis)
- **Comparative embryology** — similar stages in embryo development
- **DNA evidence** — % similarity indicates relatedness
- **Selective breeding** — artificial selection demonstrates variation and change

Speciation

DEFINITION

A new species arises when populations can no longer interbreed to produce fertile offspring.

ALLOPATRIC SPECIATION

- Geographic barrier separates population
- Gene pools diverge independently
- Different mutations accumulate → reproductive isolation
- Example: Darwin's finches (Galápagos)

SYMPATRIC SPECIATION

- No geographic barrier; occurs in same area
- Polyploidy in plants (e.g. wheat)
- Ecological/behavioural isolation

Classification & Taxonomy

BINOMIAL NOMENCLATURE

- Genus species (italicised, e.g. Homo sapiens)
- Genus capitalised; species lowercase

TAXONOMIC HIERARCHY (KING PHILIP CAME OVER FOR GOOD SOUP)

Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species

THREE DOMAINS

- **Bacteria** — prokaryotes; peptidoglycan cell wall
- **Archaea** — prokaryotes; extremophiles; different cell wall
- **Eukarya** — all eukaryotes (fungi, plants, animals, protists)

HL Cladistics groups organisms by shared derived characters; produces cladograms based on evolutionary relationships, not morphology alone

> 💡 **QUICK RECALL:** A species = group of organisms that can interbreed to produce FERTILE offspring. Mule (horse × donkey) is sterile = different species.

> ⚠ **COMMON MISTAKE:** Evolution is NOT directional or goal-oriented. Natural selection acts on EXISTING variation — it doesn't create new traits to meet needs.

Digestive System — Organs & Functions

Organ	Function	Enzymes/Secretions
Mouth	Mechanical digestion; starts starch breakdown	Salivary amylase
Oesophagus	Moves food via peristalsis	None
Stomach	Protein digestion; kills bacteria	Pepsin; HCl (pH 1.5–2)
Small intestine	Digestion + absorption	Amylase, lipase, proteases
Pancreas	Secretes digestive enzymes + $NaHCO_3$	Amylase, lipase, trypsin
Liver	Produces bile; metabolises nutrients	Bile salts
Large intestine	Water absorption; faeces formation	Bacteria; no enzymes

Absorption in Small Intestine

VILLI ADAPTATIONS

- **Villi** — finger-like projections ↑ surface area
- **Microvilli** (brush border) — further ↑ SA
- **Capillary network** — glucose and amino acids → blood → portal vein → liver
- **Lacteal** (lymph vessel) — absorbs fatty acids and glycerol (as chylomicrons)
- Thin epithelium — short diffusion distance
- Rich blood supply — maintains concentration gradient

ENZYME DIGESTION SUMMARY

- Starch → Maltose → Glucose (amylase → maltase)
- Proteins → Peptides → Amino acids (proteases → peptidases)
- Triglycerides → Fatty acids + Glycerol (lipase)

Hormonal Control of Digestion

Hormone	Source	Target	Effect
Gastrin	Stomach	Stomach	↑ HCl secretion; ↑ pepsin
Secretin	Duodenum	Pancreas	↑ $NaHCO_3$ (neutralise acid)
CCK	Duodenum	Pancreas, Gallbladder	↑ enzymes; bile release

Bile & Emulsification

- **Bile** — produced in liver, stored in gallbladder
- Contains bile salts, bile pigments (bilirubin)
- **Emulsification** — breaks large fat globules into small droplets
- ↑ surface area for lipase action
- Bile is NOT an enzyme — it does not chemically digest fats
- Bile salts also neutralise acid from stomach

> ⚠ **COMMON MISTAKE:** Bile EMULSIFIES fats — it does NOT digest them. Lipase digests fats. Don't confuse in exam!

> ⚡ **EXAM TIP:** IB commonly asks: "Explain how the structure of the small intestine is adapted for absorption." List ≥ 4 adaptations with function (villi, microvilli, capillaries, lacteal, thin epithelium).

Aerobic Cellular Respiration — Overview

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$$

1. Glycolysis

LOCATION: CYTOPLASM

- Glucose (6C) → 2 Pyruvate (3C)
- Net **2 ATP** produced
- 2 NADH produced
- No O_2 required

2. Krebs Cycle

LOCATION: MITOCHONDRIAL MATRIX

- Pyruvate → Acetyl CoA (decarboxylation)
- Per glucose: 2 ATP, 6 NADH, 2 $FADH_2$
- CO_2 released
- Requires O_2 (indirectly)

3. Electron Transport Chain

LOCATION: INNER MITOCHONDRIAL MEMBRANE

- $NADH/FADH_2$ → electrons → H^+ gradient
- **~32 ATP** produced (chemiosmosis)
- O_2 = final electron acceptor → H_2O
- Requires O_2

Total ATP from aerobic respiration ≈ 36–38 ATP per glucose

⚠ Anaerobic Respiration

IN ANIMALS (LACTIC ACID FERMENTATION)

Glucose → 2 Pyruvate → 2 Lactic acid
Net: 2 ATP only

- Occurs during intense exercise when O_2 insufficient
- Lactic acid accumulates → muscle fatigue
- "Oxygen debt" repaid when O_2 available again

IN YEAST (ALCOHOLIC FERMENTATION)

Glucose → 2 Pyruvate → 2 Ethanol + 2 CO_2
Net: 2 ATP only

Gas Exchange in Lungs

ALVEOLI ADAPTATIONS

- Large surface area (millions of alveoli)
- Moist surface — gases dissolve
- One cell thick (squamous epithelium)
- Rich capillary network — maintains concentration gradient
- O_2 diffuses into blood; CO_2 diffuses out

BREATHING MECHANICS

- **Inhale**: diaphragm + intercostals contract → ↑ volume → ↓ pressure
- **Exhale**: muscles relax → ↓ volume → ↑ pressure → air out

Aerobic respiration: Glycolysis → Pyruvate oxidation → Krebs cycle → ETC + Oxidative phosphorylation → ~38 ATP

⚠ **COMMON MISTAKE:** Glycolysis does NOT require oxygen. It occurs in the cytoplasm regardless of oxygen availability. O_2 is only required at the ETC.

💡 **QUICK RECALL:** ATP yield: Glycolysis = 2, Krebs = 2, ETC = ~32–34. Always say "approximately" — exact number debated.

SECTION 6 — CIRCULATION & IMMUNITY

Circulatory System

DOUBLE CIRCULATION

Pulmonary	Heart → Lungs → Heart (oxygenation)
Systemic	Heart → Body → Heart (delivery)

HEART STRUCTURE

- 4 chambers: 2 atria + 2 ventricles
- Left ventricle — thicker wall (pumps to body at higher pressure)
- **Valves**: atrioventricular (prevent backflow), semilunar
- Sinoatrial node (SAN) = pacemaker; initiates contraction
- Atrioventricular node (AVN) delays signal → ventricles fill

BLOOD VESSELS

Vessel	Wall	Function
Artery	Thick, muscular, elastic	Away from heart; high pressure
Vein	Thin; valves	Towards heart; low pressure
Capillary	One cell thick	Exchange with tissues

Blood Components

Component	Key Features	Function
Erythrocytes (RBC)	No nucleus; biconcave; haemoglobin	O_2 transport (oxyhaemoglobin)
Leucocytes (WBC)	Nucleated; various types	Immunity; pathogen defence
Platelets	Cell fragments; no nucleus	Blood clotting (fibrin)
Plasma	Liquid (90% water)	Transport: nutrients, hormones, CO_2, waste

HAEMOGLOBIN & O_2 TRANSPORT

- $Hb + 4O_2 \rightleftharpoons HbO_2$ (oxyhaemoglobin)
- **Bohr effect**: ↑ CO_2 → ↓ pH → lower O_2 affinity → O_2 released to tissues
- O_2 dissociation curve — sigmoid shape; cooperative binding
- Fetal Hb — higher affinity than adult Hb

Immunity

FIRST LINE OF DEFENCE

- Skin — physical barrier
- Mucus — traps pathogens
- Cilia — sweep mucus
- Stomach acid (pH 1.5–2)
- Tears/saliva — lysozyme

SECOND LINE (NON-SPECIFIC)

- **Phagocytosis** — neutrophils engulf pathogens; lysosomes destroy
- **Inflammation** — histamine release; vasodilation; ↑ blood flow
- **Fever** — ↑ temperature inhibits pathogen growth

THIRD LINE (SPECIFIC/ADAPTIVE)

- **B lymphocytes** — produce antibodies (humoral)
- **T lymphocytes** — attack infected cells (cell-mediated)
- **Antibody** — binds specific antigen; 4 polypeptides; Y-shape
- **Memory cells** — rapid response on re-exposure (immunity)

Vaccination

- Introduces antigen (dead/weakened pathogen, or antigen alone)
- Activates B and T cells → memory cells form
- Second exposure → rapid antibody response
- Herd immunity — enough vaccinated to protect community

EXAM TIP: IB regularly asks to "outline the process of antibody production." Include: antigen presentation → T helper activation → B cell activation → plasma cells → antibody secretion → memory cells.

COMMON MISTAKE: Antibodies are produced by PLASMA CELLS (differentiated B cells), not by T cells or B cells directly.

★ HL EXTENSIONS — NUCLEIC ACIDS & METABOLISM

HL ONLY

HL Nucleic Acids — Advanced

DNA VS RNA

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Strands	Double	Single (mostly)
Location	Nucleus (+ mitochondria)	Nucleus + cytoplasm
Stability	More stable	Less stable

TYPES OF RNA

- **mRNA** — carries genetic code
- **tRNA** — brings amino acids; cloverleaf shape
- **rRNA** — ribosome structure
- **snRNA** — splicing of introns

SUPERCOILING & HISTONES

- DNA wrapped around **histone octamers** → nucleosomes
- Nucleosomes compact DNA ~7-fold
- Further coiling → chromatin → chromosomes
- Allows 2m of DNA to fit in nucleus

GENE EXPRESSION REGULATION

- **Operons** (prokaryotes) — lac operon: repressor controls transcription
- **Epigenetics** — methylation (silences), acetylation (activates)
- **siRNA** — post-transcriptional gene silencing
- Transcription factors bind promoter regions

HL Metabolism — Photosynthesis

LIGHT-DEPENDENT REACTIONS (THYLAKOID)

1. Chlorophyll absorbs light energy (Photosystem II)
2. Water photolysis: $H_2O \rightarrow O_2 + H^+ + e^-$
3. Electrons pass along ETC; H^+ gradient forms
4. ATP synthesised (photophosphorylation)
5. Photosystem I: $NADP^+ + H^+ + e^- \rightarrow NADPH$

Products: ATP, NADPH, O_2 (waste)

LIGHT-INDEPENDENT (CALVIN CYCLE — STROMA)

1. CO_2 + RuBP (5C) → 2 × GP (3C) [RuBisCO enzyme]
2. GP + ATP + NADPH → G3P (triose phosphate)
3. G3P used for glucose, amino acids, fatty acids
4. RuBP regenerated using ATP

Products: Glucose (G3P → glucose 6-phosphate → glucose)

HL Aerobic Respiration — HL Detail

Glycolysis (Cytoplasm):

- 2 ATP used, 4 ATP made → net 2 ATP
- 2 NAD^+ → 2 NADH
- Glucose-6-phosphate → pyruvate

Krebs Cycle (Matrix):

- Per turn: 1 ATP, 3 NADH, 1 $FADH_2$, $2CO_2$
- 2 turns per glucose
- Citrate, oxaloacetate key intermediates

ETC (Inner membrane):

- NADH → 2.5 ATP; $FADH_2$ → 1.5 ATP
- Chemiosmosis via ATP synthase
- $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

⭐ HL EXTENSIONS — PLANT BIOLOGY & ANIMAL PHYSIOLOGY

HL ONLY

HL Plant Biology

TRANSPORT IN PLANTS

- **Xylem** — water + mineral ions (UPWARD); dead cells; no end walls
- **Phloem** — sugar (sucrose) in both directions; living sieve tube cells
- **Transpiration** — evaporation from leaves pulls water column up (cohesion-tension)
- **Cohesion** — water molecules attract each other (H-bonds)
- **Adhesion** — water attracted to xylem walls
- **Root pressure** — active loading of ions into xylem

FACTORS AFFECTING TRANSPIRATION

- ↑ Temperature → ↑ rate
- ↑ Wind → ↑ rate
- ↑ Humidity → ↓ rate
- ↑ Light → stomata open → ↑ rate

STOMATA & GUARD CELLS

- Guard cells control stomatal aperture
- In light: K^+ ions pumped in → guard cells turgid → stomata open
- In dark/drought: K^+ out → guard cells flaccid → stomata close
- ABA (abscisic acid) causes stomata to close during water stress

PHLOEM LOADING (SOURCE TO SINK)

- Active loading of sucrose into phloem at source (leaf)
- ↑ solute → water enters → ↑ pressure
- Sugar flows to sink (fruit, roots)
- Water exits at sink; sucrose unloaded

HL Hormones & Homeostasis

BLOOD GLUCOSE REGULATION

Hormone	Source	Effect
Insulin	β cells (pancreas)	↓ blood glucose (glycogen synthesis)
Glucagon	α cells (pancreas)	↑ blood glucose (glycogen → glucose)

- **Type 1 diabetes** — autoimmune destruction of β cells; no insulin produced
- **Type 2 diabetes** — cells resistant to insulin; lifestyle factors

KIDNEY & OSMOREGULATION

- Nephron unit: glomerulus → Bowman's capsule → PCT → loop of Henle → DCT → collecting duct
- **Ultrafiltration** at glomerulus (high pressure)
- **Selective reabsorption**: glucose, amino acids, Na^+ (PCT)
- **ADH** (antidiuretic hormone) — ↑ water reabsorption in collecting duct
- ↑ blood osmolarity → hypothalamus → ↑ ADH → concentrated urine

HL Neuroscience Basics

- **Resting potential**: -70 mV (inside negative; Na^+/K^+ pump)
- **Action potential**: Na^+ channels open → depolarisation (+30 mV) → K^+ channels open → repolarisation
- **All-or-nothing principle** — threshold must be reached
- **Refractory period** — ensures unidirectional propagation
- **Myelination** — saltatory conduction; faster signal
- **Synapse**: presynaptic vesicles release neurotransmitters → bind postsynaptic receptors → new action potential
- **Acetylcholine** — common neurotransmitter; broken down by acetylcholinesterase
- **Serotonin** — mood regulation; SSRIs block reuptake

⚠ EXAM MASTERY — 15 MOST COMMON BIOLOGY MISTAKES

❌ Cell Biology Mistakes

- M1 **Ribosomes are organelles** — FALSE. They are not membrane-bound.
- M2 **Osmosis = diffusion** — No. Osmosis is WATER only, via semipermeable membrane.
- M3 **Active transport doesn't need a gradient** — Correct, but students forget it requires ATP + carrier proteins.
- M4 **Cell wall = cell membrane** — Cell wall is OUTSIDE the membrane; different structures.

❌ Molecular Biology Mistakes

- M5 **tRNA has the codon** — FALSE. tRNA has the ANTICODON. mRNA has codons.
- M6 **Enzymes are "used up"** — Enzymes are not consumed; they are reused after each reaction.
- M7 **DNA polymerase reads 5'→3'** — DNA pol reads template 3'→5' and synthesises 5'→3'.
- M8 **Transcription occurs in cytoplasm** — In eukaryotes, transcription occurs in the NUCLEUS.

❌ Genetics Mistakes

- M9 **Mutation always harmful** — Mutations can be neutral, beneficial, or harmful.
- M10 **Writing sex-linked crosses without X/Y notation** — Always write X^N not just "N" for sex-linked alleles.
- M11 **Meiosis = mitosis** — Meiosis: 4 haploid cells, genetic variation. Mitosis: 2 diploid, identical cells.

❌ Physiology Mistakes

- M12 **Bile digests fat** — Bile EMULSIFIES fats. Lipase digests them. Bile ≠ enzyme.
- M13 **Glycolysis needs O_2** — Glycolysis is anaerobic; occurs in cytoplasm regardless of oxygen.
- M14 **Antibodies made by T cells** — Antibodies are produced by PLASMA CELLS (differentiated B lymphocytes).
- M15 **Arteries always carry oxygenated blood** — Pulmonary artery carries DEoxygenated blood to lungs.

⚠ Evolution & Ecology Mistakes

- **Evolution is goal-directed** — False; natural selection acts on existing variation without direction.
- **"Survival of the fittest" = strongest** — "Fittest" means best adapted to environment, not physically strongest.
- **Energy is recycled in ecosystems** — Energy FLOWS (not recycled); only matter/nutrients cycle.

✅ IB Mark Scheme Precision

✓ Say "semi-conservative replication" not just "replication"

✓ Include units in all numerical answers

✓ Use specific enzyme names (e.g., DNA polymerase III)

✓ Use "concentration gradient" not just "gradient"

✓ State direction of movement (e.g., "from high to low")

✓ Name specific bacteria in nitrogen cycle

✓ Distinguish between "describe" and "explain"

EXAM STRATEGY & IB COMMAND TERMS

Paper-by-Paper Strategy

PAPER 1 — MCQ (SL: 30 QS / HL: 40 QS)

- Eliminate obviously wrong answers first
- Watch for "always," "never," "only" — usually wrong
- Re-read question if >2 answers seem correct
- Never leave blank — guess if unsure

PAPER 2 — SHORT & LONG ANSWER

- Check how many marks per question — match answer depth
- 1 mark = 1 distinct point
- Draw large, clear, labeled diagrams
- Use biological terminology precisely

PAPER 3 — DATA-BASED & OPTIONS

- Describe trends before explaining causes
- Quote data (with units) to support claims
- State anomalies if present

IB Command Terms — Biology

Term	What to do	Detail
State	Give a fact	No explanation needed; one sentence
Define	Give exact meaning	Use precise biological language
Outline	Brief account	Main points only; no full explanation
Describe	Detail features/processes	What happens — no why
Explain	Give reasons/mechanisms	Why it happens — cause & effect
Compare	Similarities AND differences	Both must be addressed
Distinguish	Differences only	Use contrast language
Analyse	Break down; interpret	Look at components & relationships
Evaluate	Assess evidence	Weigh pros/cons; reach conclusion
Suggest	Propose a hypothesis	Reasoned answer; no definite answer needed
Calculate	Show working numerically	Show all steps; include units
Construct	Draw/design	Labeled, to scale if needed
Annotate	Add notes to diagram	Labels with brief explanations

Time Management in Exams

ALLOCATE TIME BY MARKS

- ~1 min per mark
- 5-mark question → ~5 min
- Leave 10% time for review

ANSWER STRUCTURE

- Write in bullet points (easier to mark)
- Underline key terms
- Include both structure AND function

DIAGRAM RULES

- Pencil for diagrams
- Label with lines, not arrows
- Neat = more marks

> ⚡ **EXAM TIP:** For 6+ mark "explain" questions: introduce the concept, give the mechanism step-by-step, explain why each step matters. Link cause to effect throughout.

📖 KEY DEFINITIONS SUMMARY — IB BIOLOGY

Cell & Molecular

Cell theory	All organisms are made of cells; cells are basic unit; cells from pre-existing cells
Osmosis	Diffusion of water across semipermeable membrane from high to low water potential
Enzyme	Biological catalyst; protein that lowers activation energy of a specific reaction
Denaturation	Irreversible change in enzyme's 3D shape; active site destroyed; not loss of function temporarily
Active site	Specific region on enzyme where substrate binds; complementary shape to substrate
Transcription	Process by which mRNA is synthesised from a DNA template using RNA polymerase
Translation	Process by which mRNA codons are decoded at ribosome to produce a polypeptide
Codon	A sequence of three mRNA bases that codes for a specific amino acid (or stop signal)

Genetics

Gene	Sequence of DNA nucleotides that codes for a functional product (protein or RNA)
Allele	One of the specific forms of a gene at a given locus; differs from other alleles by one or more bases
Codominance	Both alleles are fully expressed in the phenotype of the heterozygote
Mutation	A heritable change in the nucleotide sequence of DNA
Meiosis	Cell division producing 4 genetically unique haploid daughter cells; occurs in gonads

Ecology & Evolution

Ecosystem	A community of living organisms interacting with their non-living physical environment
Niche	The role and position a species occupies in its environment; its resources and interactions
Natural selection	Differential survival and reproduction of individuals with heritable advantageous traits
Species	A group of organisms capable of interbreeding to produce fertile offspring
Speciation	The evolution of new species through reproductive isolation of populations
Carrying capacity (K)	Maximum population size an environment can sustain with available resources

Human Physiology

Peristalsis	Wave-like muscular contractions that move food along the alimentary canal
Emulsification	Breaking large fat globules into smaller droplets to increase surface area for lipase
Antigen	A molecule (usually protein) on a pathogen's surface that triggers an immune response
Antibody	A Y-shaped glycoprotein produced by plasma cells that binds to a specific antigen
Homeostasis	The maintenance of constant internal conditions in the body (e.g. temperature, glucose)
Bohr effect	Decrease in Hb's O_2 affinity due to decreased pH caused by increased CO_2 in tissues

💡 **QUICK RECALL:** Learn definitions WORD-FOR-WORD — IB mark schemes are precise. Paraphrasing often loses marks. Use the exact terms above.

QUICK REVISION — FULL SYLLABUS BULLETS

Cell Biology

- Prokaryotes: 70S ribosomes, circular DNA, no membrane-bound organelles
- Eukaryotes: 80S ribosomes, linear DNA, membrane-bound organelles
- SA:V decreases as cell size increases → cell must divide
- Phospholipid bilayer: hydrophilic heads out, hydrophobic tails in
- Osmosis: water moves to lower water potential (more solutes)
- Active transport: against gradient, requires ATP + carrier protein
- Endocytosis/exocytosis: bulk transport, requires ATP

Molecular Biology

- Condensation = bond + water out; hydrolysis = bond break + water in
- DNA: A=T (2H), G≡C (3H), antiparallel, double helix
- Replication: semi-conservative, helicase, DNA pol, ligase
- RNA pol reads 3'→5', synthesises mRNA 5'→3'
- Translation: AUG start, ribosome moves 5'→3', tRNA brings AA
- Enzymes not consumed; lower Ea; denaturation is irreversible
- Competitive inhibition overcome by ↑ substrate; non-competitive is not

Genetics

- Monohybrid Aa × Aa: 3:1 phenotype ratio
- Test cross × aa: 1:1 if heterozygous
- Sex-linked: X chromosome carries allele; males (XY) show if even 1 recessive
- Meiosis produces 4 haploid cells; crossing over in Prophase I
- Frameshift > substitution in severity
- PCR: denature → anneal primers → extension (Taq pol)
- Gel electrophoresis: smaller fragments travel further

Ecology

- Only ~10% energy transferred between trophic levels
- Carbon fixed by photosynthesis; released by respiration/combustion
- N_2 fixed by Rhizobium; nitrified by Nitrosomonas → Nitrobacter
- Denitrification: Pseudomonas; anaerobic; N_2 released
- J-curve: exponential; S-curve: logistic (carrying capacity)
- Energy flows; matter cycles

Evolution

- Natural selection: variation → overproduction → competition → differential survival → adaptation
- Allopatric: geographic barrier; sympatric: same area
- Species = interbreed to produce FERTILE offspring
- Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species
- Homologous = same origin; analogous = same function
- HL: Cladograms based on shared derived characters

Human Physiology

- Villi/microvilli ↑ SA; capillaries → portal vein; lacteal → lymph
- Bile emulsifies fats; lipase digests them
- Glycolysis → 2 ATP; Krebs → 2 ATP; ETC → ~32 ATP
- Anaerobic: 2 ATP only; lactic acid in animals; ethanol + CO_2 in yeast
- Left ventricle thicker (pumps to body at higher pressure)
- SAN = pacemaker; AVN delays signal
- Antibodies: Y-shaped; made by plasma cells; bind specific antigens
- Memory cells → faster secondary response
- Bohr effect: ↑ CO_2 → ↓ Hb-O_2 affinity → O_2 released to tissues

Formulas to Remember

- **SA:V ratio** — SA = 6×side², V = side³, Ratio = 6/side
- **Glucose formula** — $C_6H_{12}O_6$

HL Quick Hits

- Xylem: dead cells, cohesion-tension pulls water up
- Phloem: active loading; source → sink

- **Photosynthesis**: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
- **Aerobic respiration**: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP

- Insulin: β cells; glucagon: α cells (pancreas)
- ADH: ↑ water reabsorption in collecting duct
- Action potential: -70 mV → +30 mV → repolarisation
- Calvin cycle: CO_2 + RuBP → GP → G3P (uses ATP + NADPH)

NIGHT BEFORE THE EXAM

Final Exam Checklist

Tick each box as you confirm readiness

📚 Knowledge — Core Topics

- [ ] I can state all 3 tenets of cell theory
- [ ] I can describe fluid mosaic model features
- [ ] I can explain semi-conservative DNA replication step-by-step
- [ ] I can outline transcription and translation processes
- [ ] I can complete any Punnett square correctly with notation
- [ ] I can determine inheritance pattern from a pedigree
- [ ] I can explain the mechanism of natural selection (6 steps)
- [ ] I know all steps of aerobic respiration with locations
- [ ] I can list 4 villi adaptations for absorption
- [ ] I can outline the specific immune response (antibody production)
- [ ] I know specific bacteria in the nitrogen cycle
- [ ] I can draw a labeled diagram of a named cell (animal/plant)

✏️ Exam Skills

- [ ] I know the difference between "describe" and "explain"
- [ ] I understand what "outline" and "state" require
- [ ] I know to quote data (with units) in Paper 3
- [ ] I will allocate ~1 minute per mark
- [ ] I know to use pencil for all diagrams
- [ ] I will not leave any MCQ blank
- [ ] I know how to analyse a graph (describe trend → suggest reason)
- [ ] I can write a lab design (variables, method, safety)

🚫 Key Mistakes I Will Avoid

- [ ] Bile EMULSIFIES fat — lipase DIGESTS fat
- [ ] Antibodies made by PLASMA CELLS, not T cells
- [ ] tRNA has ANTICODON, mRNA has CODON
- [ ] Glycolysis does NOT require oxygen
- [ ] Write X^N notation for sex-linked alleles

YOU'VE GOT THIS

Tomorrow, show what you know.

✦ Sleep well tonight — memory consolidates during sleep
✦ Eat a good breakfast — your brain needs glucose
✦ Read every question twice before answering
✦ Trust your preparation. You are ready.

GOOD LUCK — AIM FOR A 7 🎯